



Mark Scheme (Results)

January 2024

Pearson BTEC Nationals in
Applied Science (31617H/1B)
Unit 1: Principles and Applications of Science I
Biology

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Unit 1: Principles and Applications of Science I Biology

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1 (a)	D plasma membrane		1
1 (b)	protein synthesis/translation	accept a description	1
1 (c)	A amyloplasts		1
1 (d)	Award one mark for an identification point and one mark for an expansion point up to a maximum of two marks. Any of the following could be an identification point or an expansion point depending on how the learner shapes their response. low magnification/resolution (1) due to/limited by wavelength of light (1) unable to view {detail (of internal cell structures)/some organelles} (1)	accept named examples	2
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
2 (a)(i)	<u>red blood</u> (cell)/erythrocyte		1
2 (a)(ii)	Award one mark for an identification point and one mark for an expansion point up to a maximum of two marks. Any of the following could be an identification point or an expansion point depending on how the learner shapes their response. allows for {smooth/even} blood flow (1) reduces friction/ prevents blood clots (1)	accept keeps blood from contact with connective tissue accept prevent blockages accept produces anti-clotting factors	2
2 (a)(iii)	Atherosclerosis develops slowly with a build-up of cholesterol, fat, foam cells and other substances in the blood forming plaque/atheroma/fatty streaks (1) in the wall, under the endothelium. The build-up causes the lumen to narrow. The narrowing of the lumen increases the blood pressure (1). This can cause more damage. This can lead to atherosclerosis. Lifestyle choices, such as a poor diet, smoking/consuming alcohol (1) or lack of exercise can increase the risk of developing atherosclerosis.	Words must be in the correct order in the paragraph ignore "drinking" unqualified	3
Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
3 (a)	(Schwann cell) nucleus	accept nuclei	1
3 (b)	sodium / Na (1) potassium / K (1)	accept in either order ignore charges on ion reject any other ion	2
3 (c)(i)	difference between values (1) $40 - 2$ substitution (1) difference divided by original X 100 $38 \div 2 \times 100$ evaluation (1) 1900 (%) OR alternative method substitution (1) final $\div$ original x 100 $40 \div 2 \times 100$ subtract 100 for percentage increase (1) -100 evaluation (1) 1900(%)	1900(%) gains three marks with or without working ECF for evaluation	3

3 (c)(ii)	Any one from: - temperature (axon) diameter	accept named medicine/ drug	1
Total = 7 marks			

Question Number	Answer	Additional Guidance	Mark
4 (a)	A description including any two from: increases chance of survival (1) resistance to /protection from chemicals (1) prevents phagocytosis (1) supports adhesion to surfaces (1) protects from white blood cells (1) can act as a source of nutrients/energy (in times of starvation) (1) accept any other valid response.	accept named white blood cell	2

4 (b)	Award one mark for an identification point and up to three marks for expansion points up to a maximum of four marks. Any of the following could be an identification point or an expansion point depending on how the learner shapes their response. (plasmids) contain DNA/genes (1) (plasmids) contain alleles for antibiotic resistance (1) gene transfer can occur (between individual cells of the same or different species) (1) can lead to bacterial pathogens with resistance to {multiple} antibiotics (1) reduces/avoids destruction from immune response/white blood cells (1) (plasmids) may code for toxins or virulence factors (1) (plasmids) may give a pathogen the ability to digest particular substances (1) Accept any other valid response.	Ignore any references to any artificial genetic modifications/genetic engineering	4
Total = 6 marks			

Question Number	Indicative content
5	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some of the indicative content, but learners should be rewarded for other relevant answers.</i> Structure The root hair cell is roughly rectangular in shape with a protrusion extending away from the root into the soil It has the following cellular components: • a cell wall • a partially permeable cell membrane • a large central vacuole • a nucleus • cytoplasm • mitochondria Function/specialisation (root hair cells) – aid in expanding the reach of plants underground – increase the surface area of the plant in contact with soil for rapid uptake of water and dissolved minerals/ions. – ensure a plant's optimal strength and health to grow cell wall – fully permeable – thinned – facilitates the uptake of water by osmosis large permanent central vacuole – maximum water and salt retention – actively transporting ions into the vacuole, lowers water potential and causes water to enter by osmosis mitochondria – many mitochondria close to the cell membrane – ATP production/energy release for active transport of mineral ions lack of chloroplasts – do not absorb sunlight as underground and so do not photosynthesise large surface area for absorption – beneficial in water and mineral ion absorption from the soil – maximises efficiency of osmosis/diffusion/active transport – more membrane for protein channels/carriers storage – contains large vacuoles for maximum storage of water and nutrients – helps with preservation Root hair cells absorb oxygen from the soil (which) is used in (aerobic) respiration in mitochondria Root hair cells produce specific enzymes (to) enable the uptake of particular minerals/ions

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes.

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts/concepts with generalised comments made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific knowledge facts/concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear, coherent and logical.
Total = 6 marks		



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